Track 1 of Modeling Workshop: Tutorials

1. **Intro to Energy Systems Optimization Models and MACRO**

   - Anna Li (Princeton)

2. **Supply-side input development: Technology Parameterization for MACRO**

   - Hongxi Luo (Princeton)

3. **MACRO Installation and Demo**

   - Luca Bonaldo (Princeton)

Intro to Energy System Optimization Models

1. Brief intro to optimization
2. How is optimization applied to energy systems?
3. What are the inputs and outputs of MACRO?

Brief Intro to Optimization

What is an optimization problem?

minimize/maximize	Objective function	(Some function that gives a value that should be minimized or maximized)
subject to	Constraints that define the feasible space	

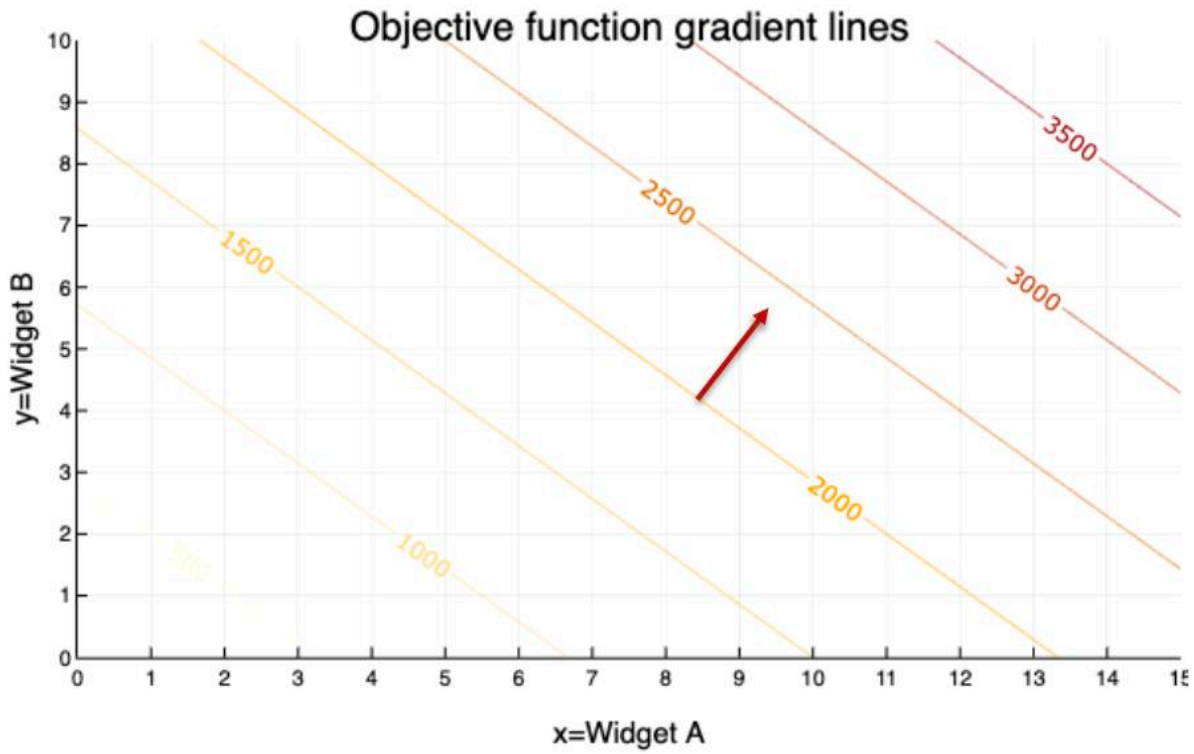
Brief Intro to Optimization

Example of a problem to maximize profit from producing two widgets

$$\max_{x,y} \; f(x,y) \qquad 150\,x + 175\,y$$

Decision variable to produce widget A

Decision variable to produce widget B



Brief Intro to Optimization

Example of a problem to maximize profit from producing two widgets

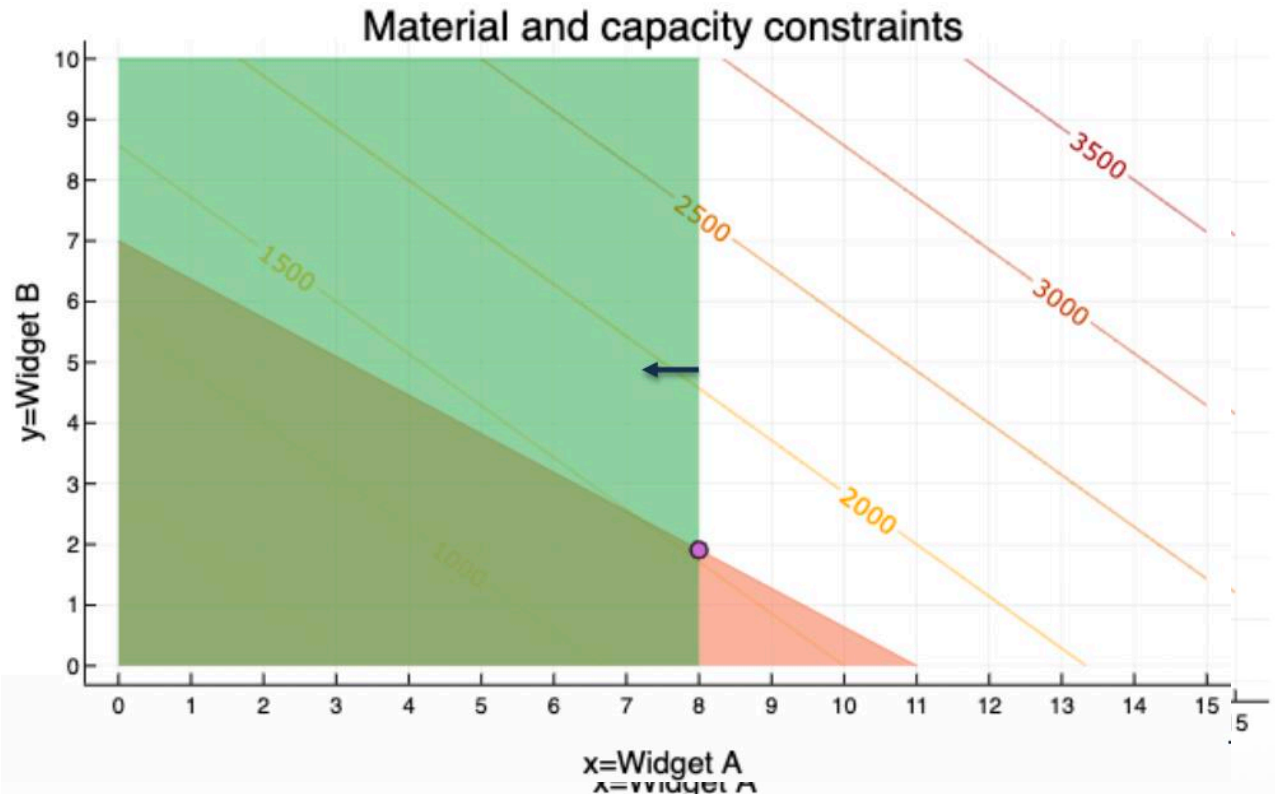
max $\underset{x,y}{} f(x,y)$ $\quad 150\,x + 175\,y$

subject to $\quad x \geq 0$

$y \geq 0$

$7\,x + 11\,y \leq 77$

$x \leq 8$



Optimization Applied to Energy Systems

What is the **objective function** in an energy system optimization model?

minimize Total System Cost = Investment Costs + Operational Costs

Optimization Applied to Energy Systems

What is the **objective function** in an energy system optimization model?

minimize Total System Cost = Investment Costs + Operational Costs

↓

Cost of building Technology 1
+
Cost of building Technology 2
+
Cost of building Technology 3

↓

How much is built (MW, tons/hour)

How much it costs to build per MW or tons/hour

Capacity of Technology 1 × Investment Cost of Technology 1
+
Capacity of Technology 2 × Investment Cost of Technology 2
+
Capacity of Technology 3 × Investment Cost of Technology 3

Decision Variable

Input Parameter

Optimization Applied to Energy Systems

What is the **objective function** in an energy system optimization model?

minimize Total System Cost = Investment Costs + Operational Costs

↓

Cost of operating Technology 1
+
Cost of operating Technology 2
+
Cost of operating Technology 3

↓

How much is produced / dispatched at time t (MWh, tons)

How much it costs to produce per MWh or ton

$$\sum_t (\text{Flow of Technology 1 at time t} \times \text{Operating Cost of Technology 1})$$
+
$$\sum_t (\text{Flow of Technology 2 at time t} \times \text{Operating Cost of Technology 2})$$
+
$$\sum_t (\text{Flow of Technology 3 at time t} \times \text{Operating Cost of Technology 3})$$

Optimization Applied to Energy Systems

What is the **objective function** in an energy system optimization model?

minimize Total System Cost = Investment Costs + Operational Costs

↓

Cost of operating Technology 1
+
Cost of operating Technology 2
+
Cost of operating Technology 3

↓

How much is produced / dispatched at time t (MWh, tons)

How much it costs to produce per MWh or ton

$$\sum_{t} (\text{Flow of Technology 1 at time t} \times \text{Operating Cost of Technology 1}) + \dots$$

Decision Variable

Input Parameter

Optimization Applied to Energy Systems

What are the **constraints** in an energy system optimization model?

minimize Total System Cost = Investment Costs + Operational Costs

subject to

1) Max Production Constraints:

   Flow of Technology n at time t $\leq$ Capacity of Technology n, for all t

   You can't produce more than what you've built for

2) Demand Constraints:

   $\sum_{n}$ (Dispatch of Technology n at time t) = Demand at time t, for all t

   You need to meet demand every hour

3) Supply Constraints:

   Dispatch of Technology n at time t $\leq a \times$ Supply at time t

   You can't use more than your supply

4) CO_2 Cap Constraints:

   $\sum_{n}\sum_{t}$ (CO_2 Emissions of Technology n at time t) $\leq$ CO_2 Cap

   You can't emit more than a certain amount of CO_2

Optimization Applied to Energy Systems

Inputs

Investment and Operational Costs

Supply and Demand Constraints

Energy System Optimization Models

minimize Total System Costs

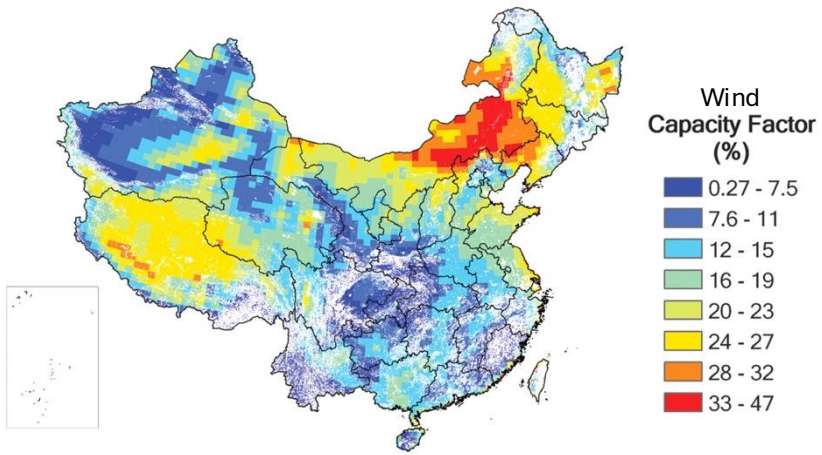
By changing decision variables

subject to System Constraints

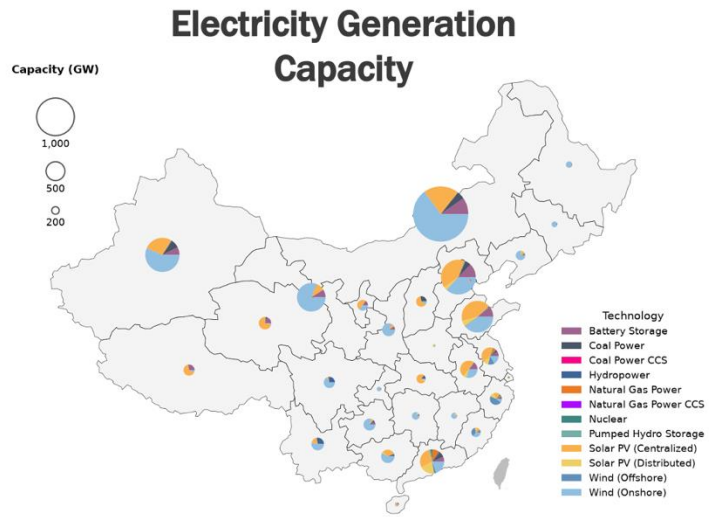
Outputs

Capacity Decisions

Operational Decisions



Michael B. McElroy *et al.* Potential for Wind-Generated Electricity in China. *Science* **325**, 1378-1380(2009). DOI: 10.1126/science.1175706



MACRO Input Structure

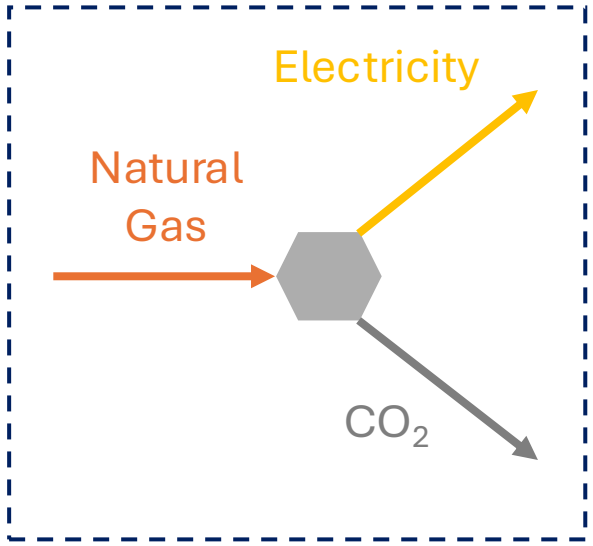
How do you represent technologies and supply/demand dynamics in MACRO?

MACRO Input Structure

How do you represent technologies and supply/demand dynamics in MACRO?

1. Technologies are modeled through their inputs and outputs, called **commodities**
    a) **Commodities** include energy, materials, and emissions (e.g., electricity, fuels, CO_2, industrial products)

MACRO Natural Gas Generator **Asset**



Assets are parameterized by ratios between inputs and outputs, and investment/operational costs (more from Hongxi later)

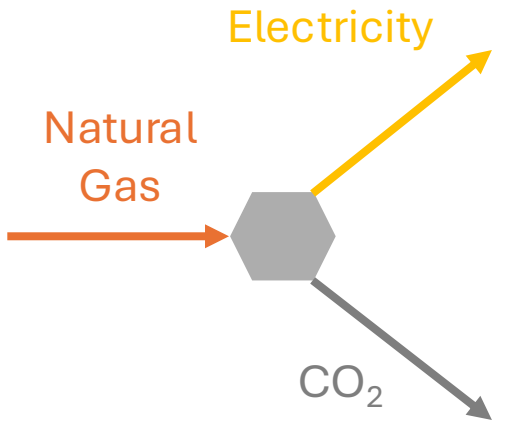
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2. Commodities flow to and from **vertices**
    a) Types of vertices

MACRO Natural Gas Generator **Asset**



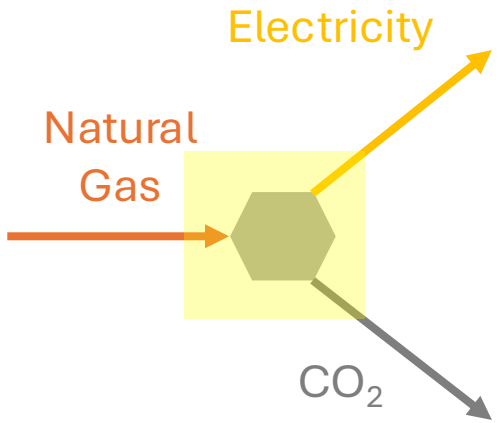
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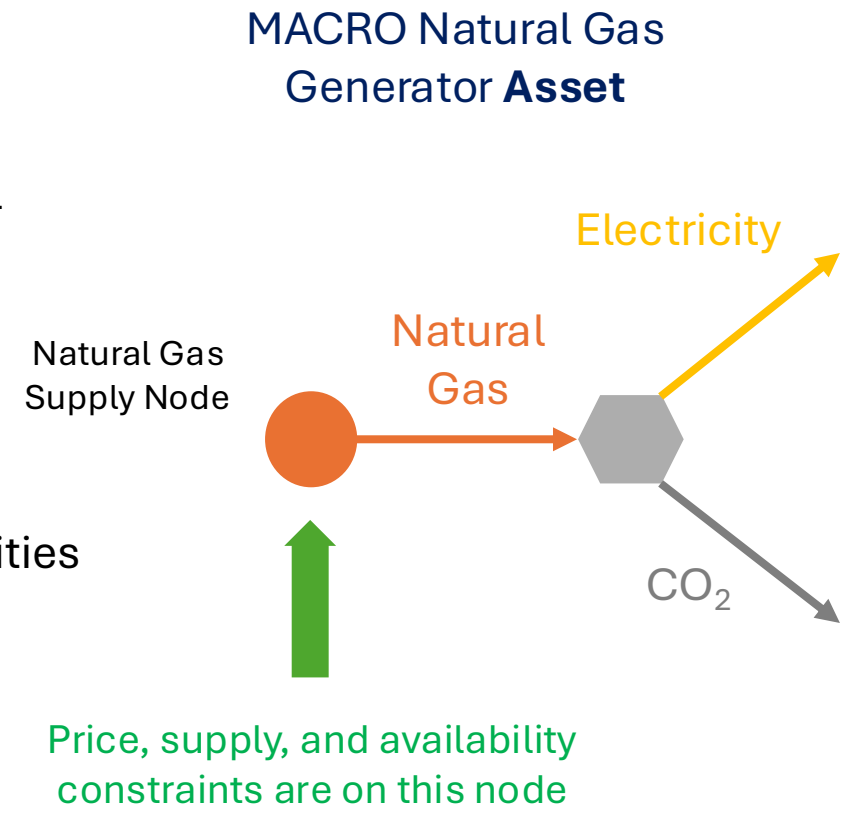


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        2) **Supply/Demand nodes** provide/consume commodities

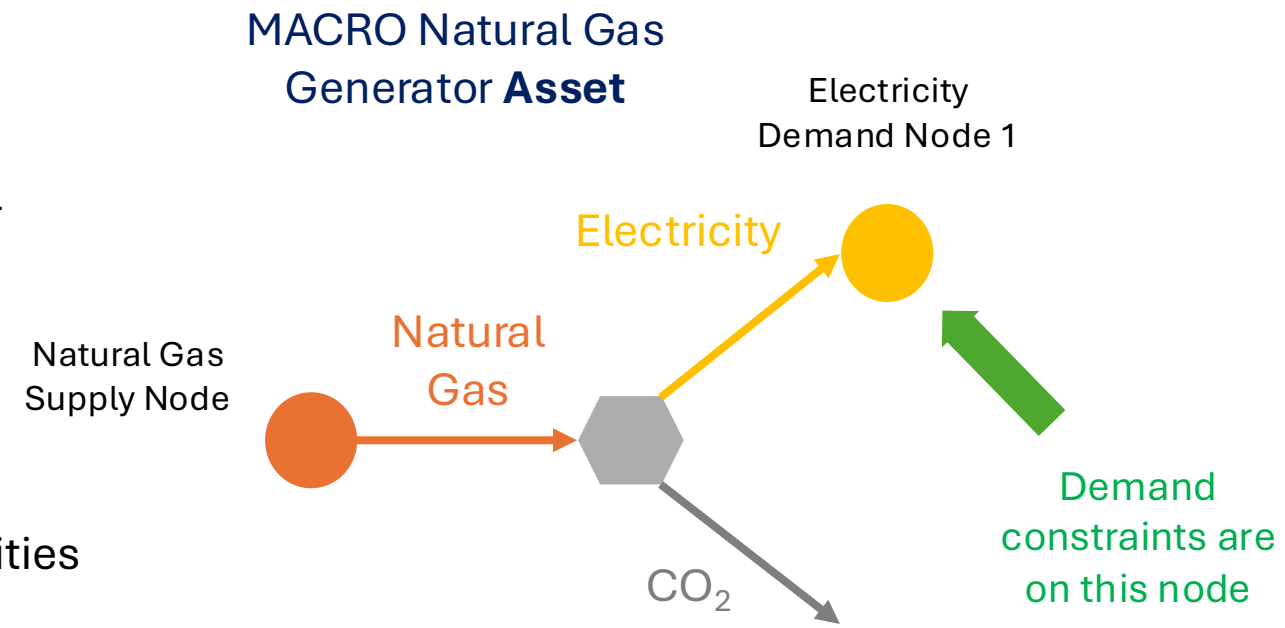


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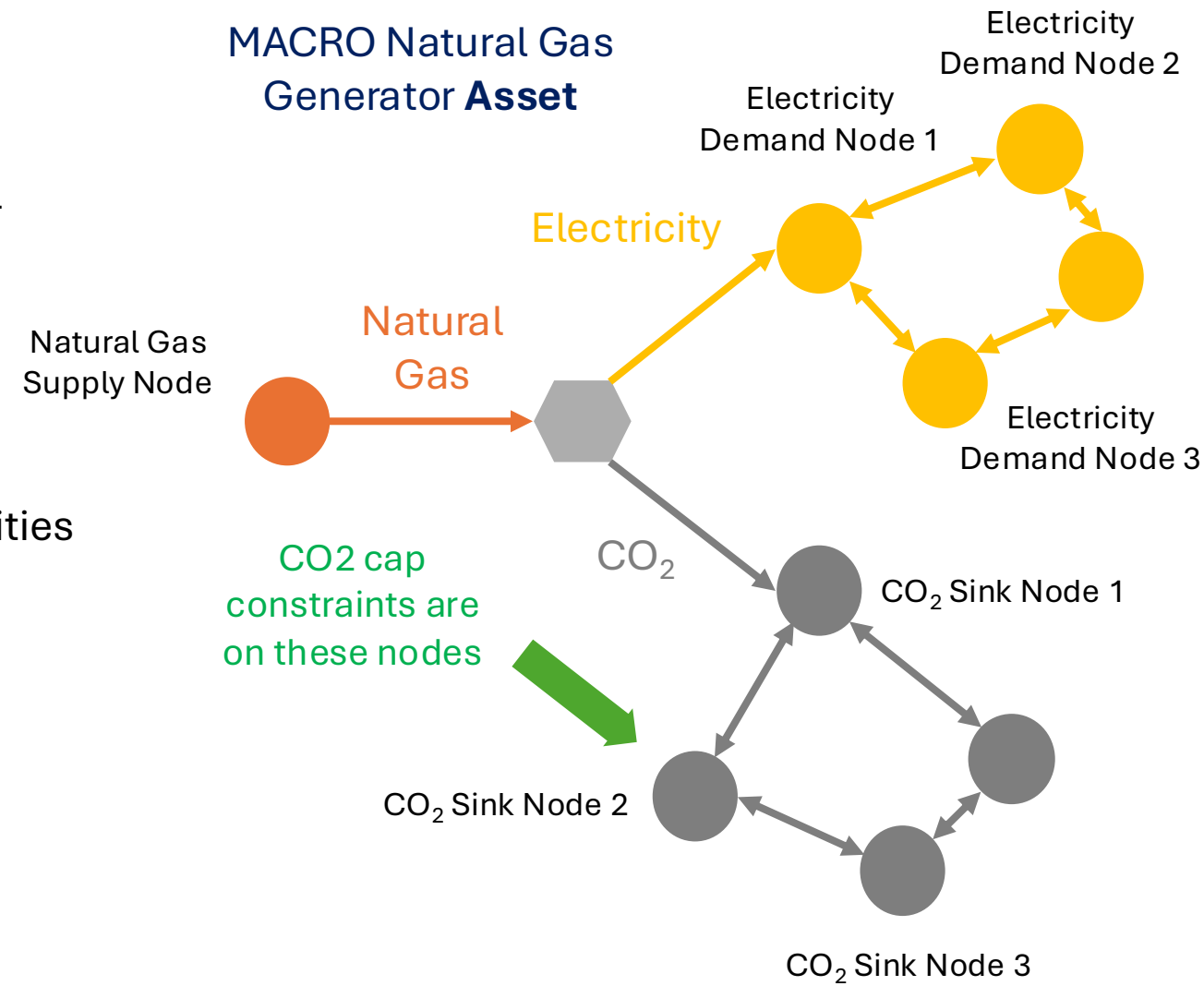
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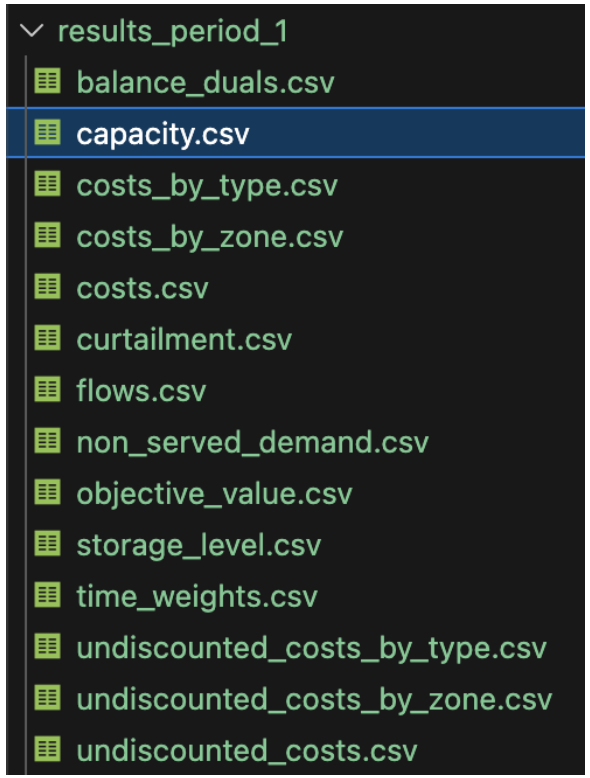
2. Commodities flow to and from **vertices**
    a) Types of vertices
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3. Vertices are connected by **edges** and carry commodity flows

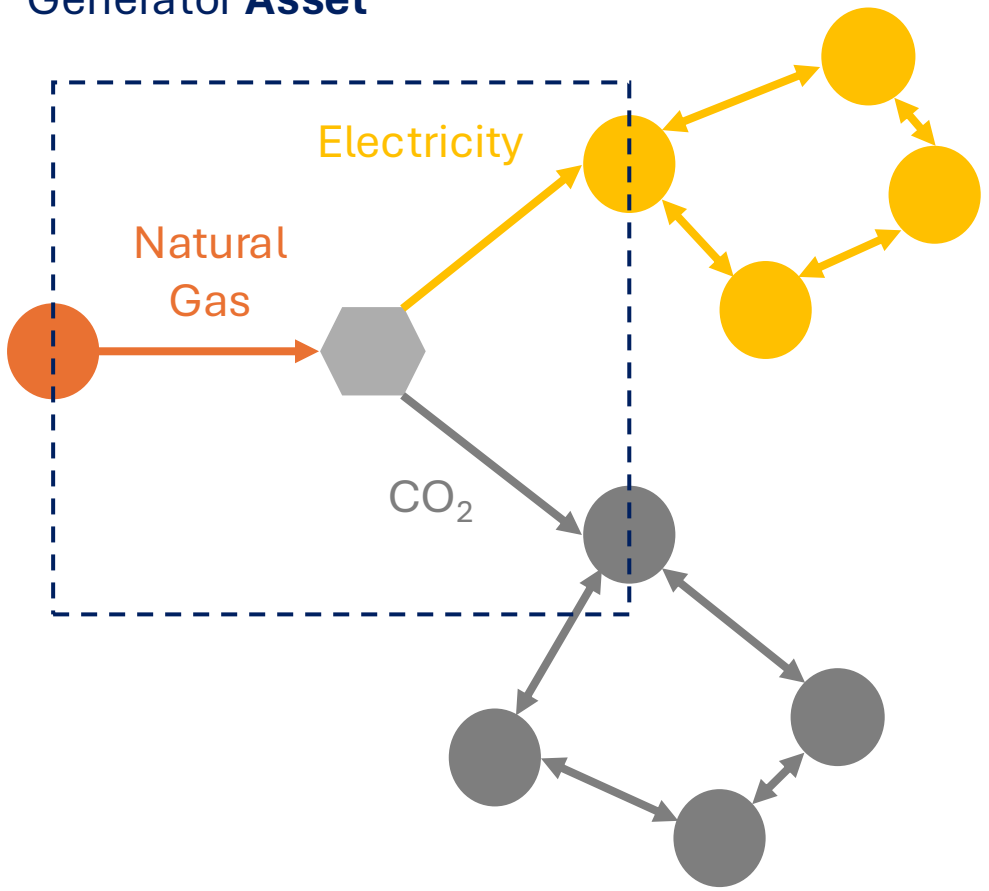


Basic MACRO Outputs

Each **asset** in MACRO has a **capacity decision**, which can be found in the "capacity.csv" file

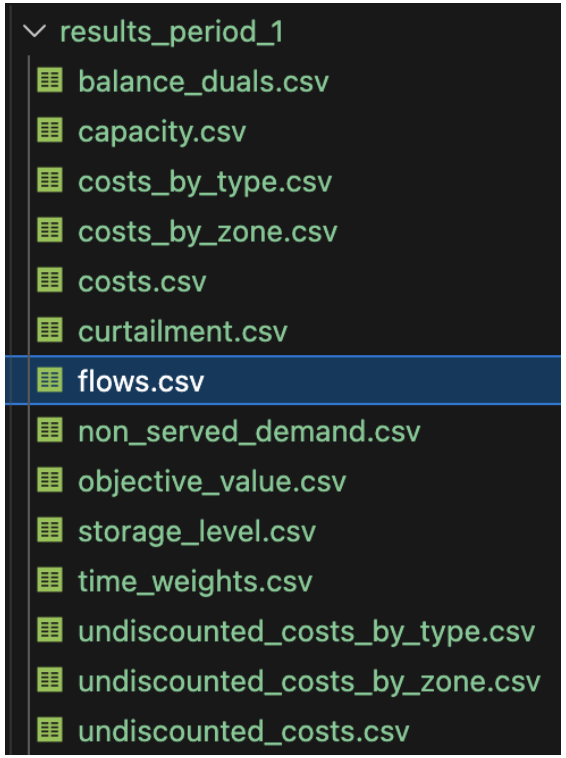


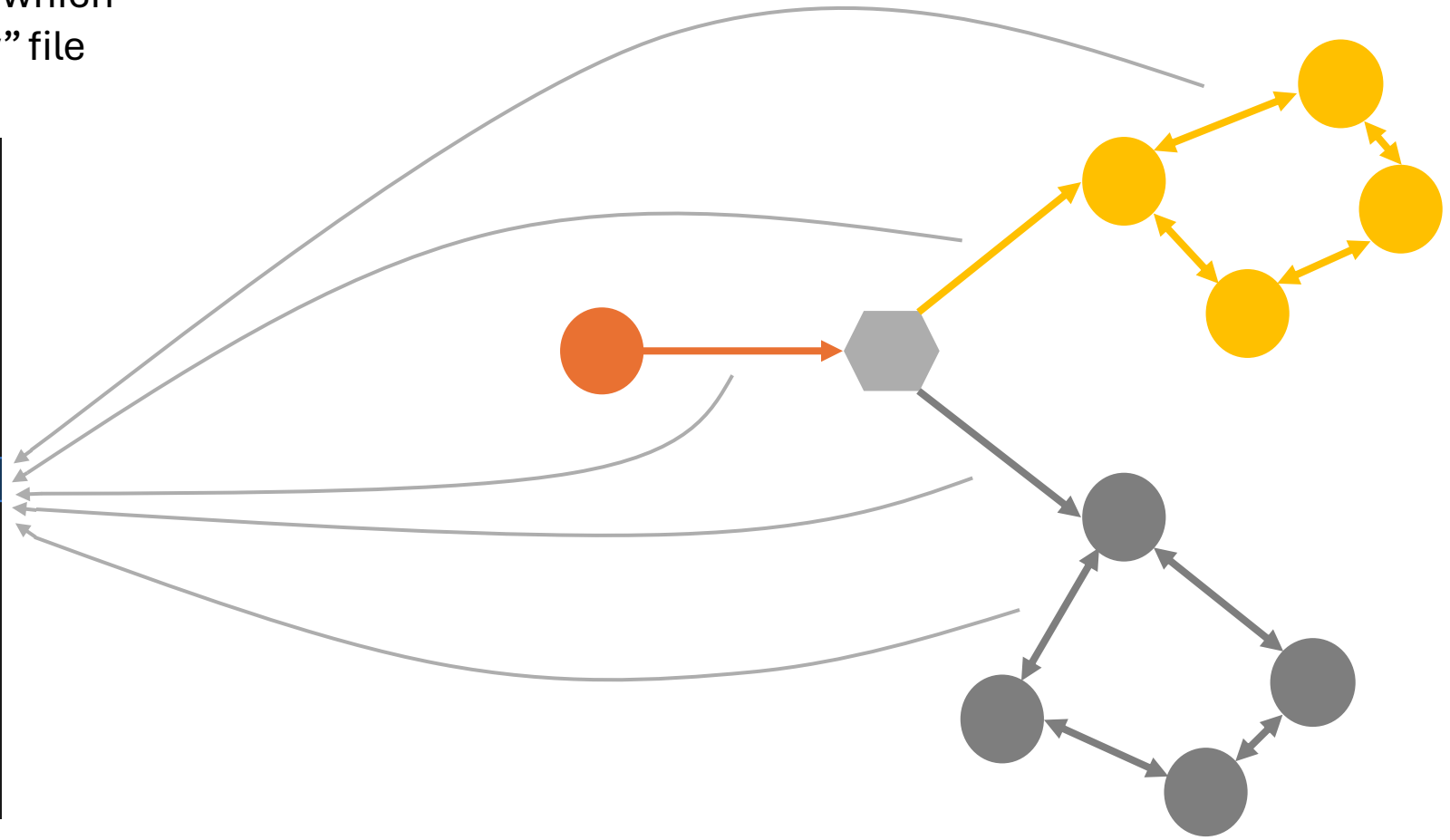
MACRO Natural Gas Generator **Asset**



Basic MACRO Outputs

Each **edge** in MACRO has an **operational decision for each timestep** of the optimization, which can be found in the “flows.csv” file





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ATR-CCS parameterization in MACRO

Mass & energy balance

Conservation laws + thermodynamics

Fuel and electricity inputs
H_2 output and CO_2 flows

Similar process and operating conditions:
MEB often varies less by region than costs.

Investment & O&M

Plant scope + local costs

Capital and fixed O&M per capacity
Variable O&M per output

Year, region and plant scale
can materially change the estimate.

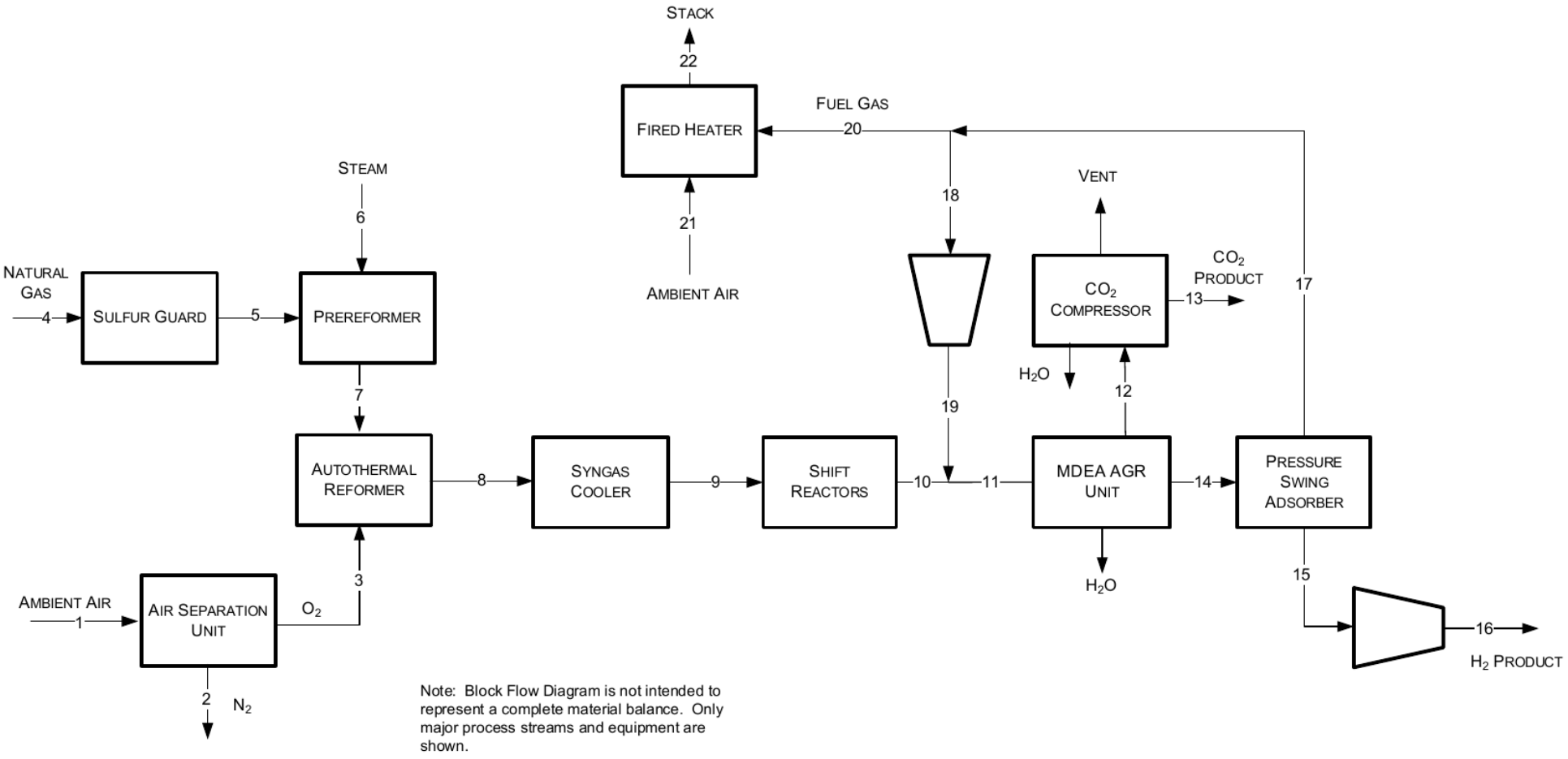
Supporting assumptions

Physical life: retirement
Economic life: capital recovery

WACC: financing
Operating limits: feasible dispatch

Sources: MACRO Thermal Hydrogen manual and TEA appendix. MEB = mass and energy balance.

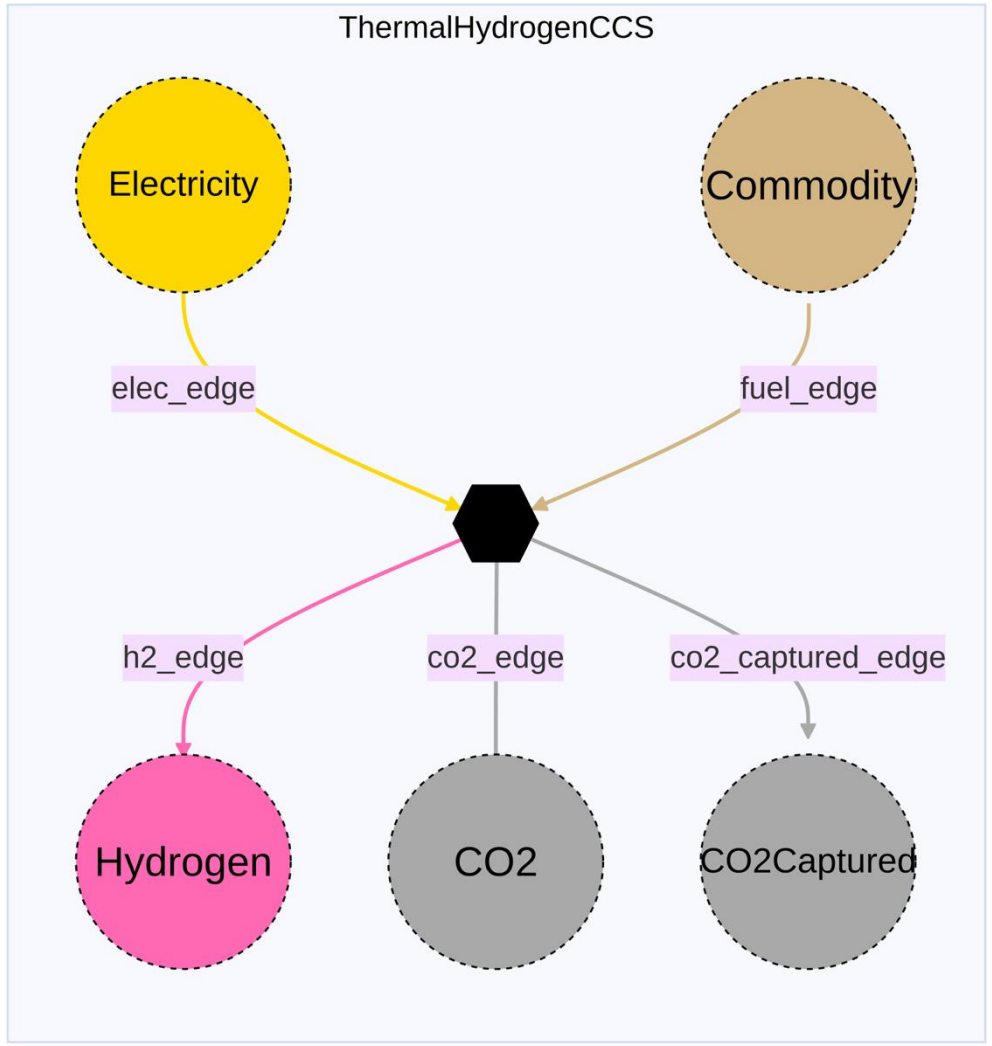
NETL Case 3: the ATR-CCS plant boundary



Air separation, reforming, H_2 purification and CO_2 compression sit inside the plant.

Source: NETL (2022), DOE/NETL-2022/3241, Exhibit 3-35, p. 102. osti.gov/servlets/purl/1862910

ThermalHydrogenCCS: one asset, five flows



Commodity = natural gas

For the ATR-CCS instance

h2_edge

Capacity and asset costs

Internal equipment

Aggregate conversion coefficients

Gas and electricity supply, and CO_2 transport/storage connect externally.

Source: tutorial_3_macro_showcase.ipynb, mermaid_diagram(ThermalHydrogenCCS). Static rendering of the notebook diagram.

HHV efficiency sets the fuel input coefficient

Cold gas efficiency = H_2 energy / fuel energy ≈ 76%

fuel_consumption = 1 / efficiency ≈ 1.32

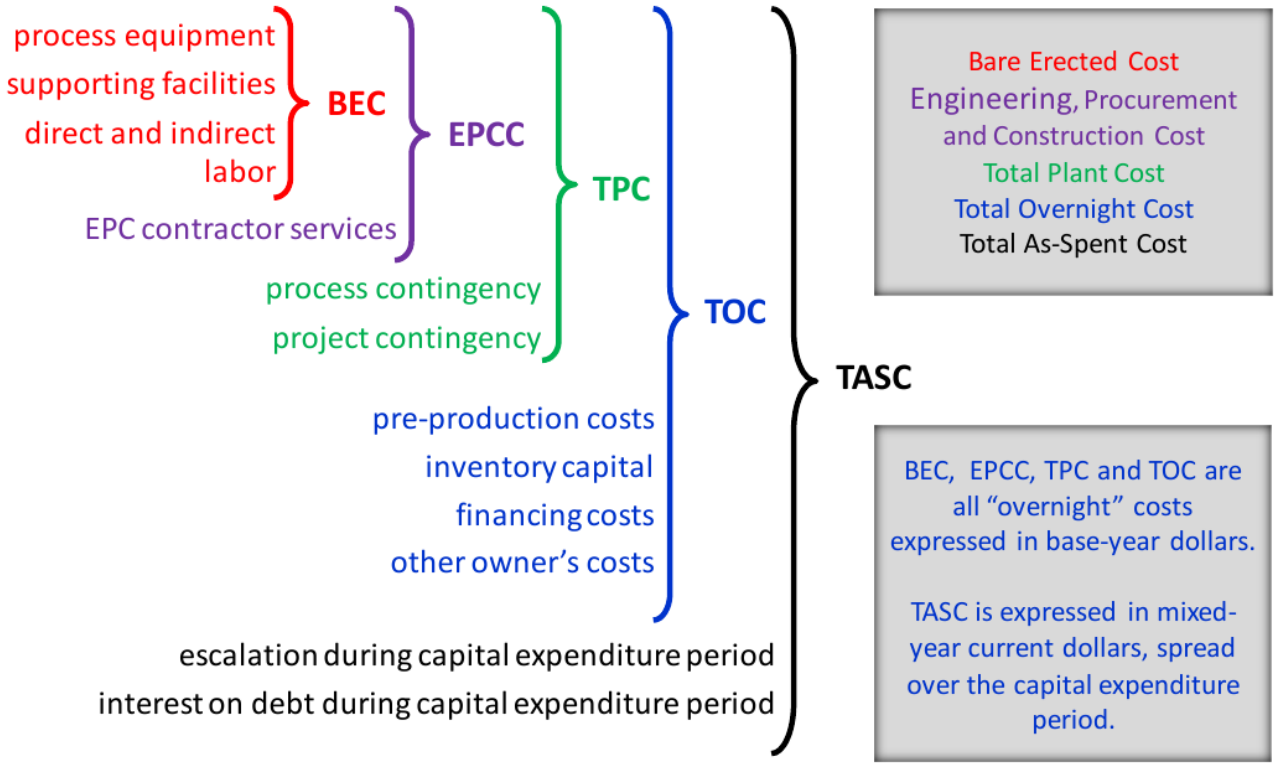
MACRO key	Value
fuel_consumption	**1.32 MWh-NG / MWh-H_2**
electricity_consumption	**0.102 MWh-electricity / MWh-H_2**
capture_rate	**0.171 t-CO_2 / MWh-NG**
emission_rate	**0.010 t-CO_2 / MWh-NG**

94.5% capture fraction corresponds to a fuel-based capture_rate.

HHV basis. H_2 output ≈ 1,066 MW. Electricity remains a separate input.

Sources: NETL Ex. 3-38, 3-39, 3-41 and 3-44. CO_2 mass = carbon mass × 44/12. Minor air/product carbon flows omitted.

ATR-CCS costs depend on scope, year and location



Cost boundary

Equipment, contingencies, owner's costs

Year and region

2018 USD, generic Midwest US site

Scale

About 1.1 GW of H_2 output

Finance

Real or nominal basis, WACC, recovery period

This example uses total overnight cost (TOC): about $905 million.

Sources: NETL Exhibit 2-23, p. 42, and site/cost assumptions. MACRO TEA appendix. TPC and TOC: base-year dollars.

ATR-CCS investment and O&M in MACRO units

TOC ≈ $905 million / H_2 capacity ≈ 1,066 MW

MACRO key	Value (2018 USD)
investment_cost	**≈ $850,000 / MW-$H_2$**
fixed_om_cost	**≈ $23,000 / MW-$H_2$-year**
variable_om_cost	**≈ $1.8 / MWh-$H_2$**

Gas, electricity and CO_2 transport/storage have separate costs.

Variable O&M here excludes purchased electricity to avoid double counting.

Sources: NETL Ex. 3-47–3-48. Current MACRO main: capital $/MW. Check the model version for annualized-cost inputs.

An illustrative LCOH of about $1.70/kg H_2

$$\text{LCOH} = \frac{\text{Annualized plant cost + annual operating costs}}{\text{Annual } H_2 \text{ production}}$$

Calculation assumptions

Real WACC 8%
Economic life 30 years
Capacity factor 90%

Annual H_2 output
≈ 217 million kg

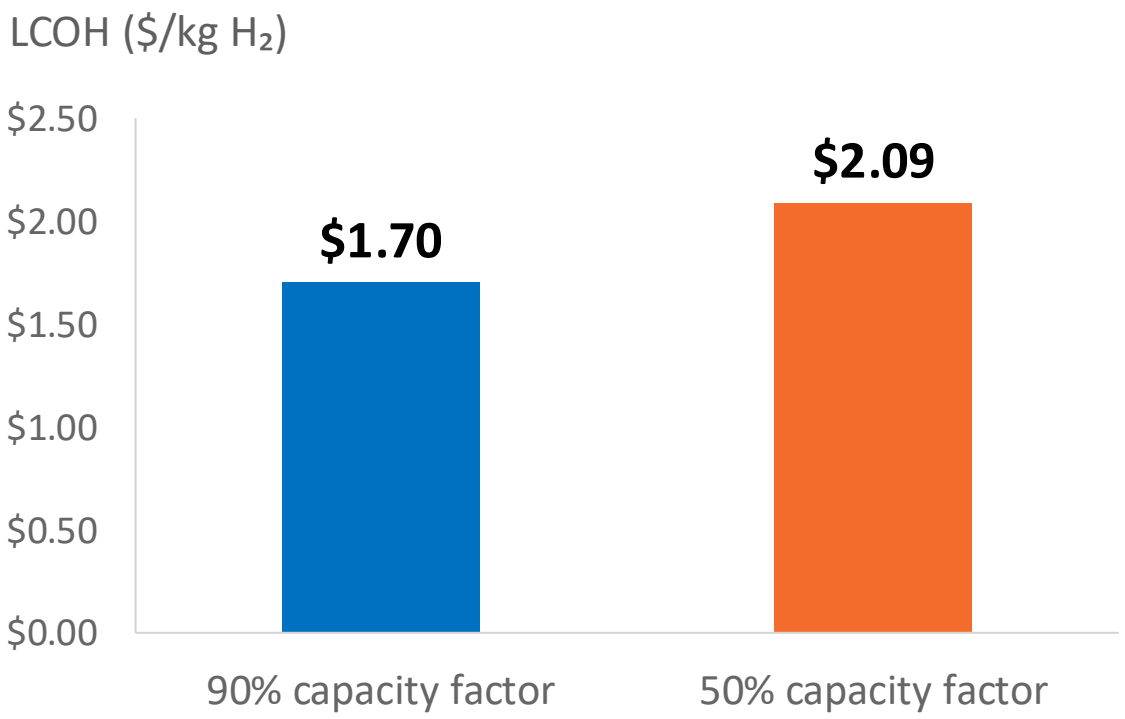
Cost component	$/kg H_2
Annualized capital	0.37
Fixed O&M	0.11
Other variable O&M	0.07
Electricity	0.29
Natural gas	0.77
CO_2 transport/storage	0.09

Energy prices: gas $4.42/MMBtu, electricity $72/MWh. CO_2 transport/storage: $10/t.

Sources: NETL Ex. 2-24 and 3-38, 3-47–3-49. NETL reports $1.59/kg with T&S under its own tax/finance method.

LCOH depends on utilization and system conditions

Same plant, different utilization



Less output spreads fixed costs over fewer kilograms.

Useful for cost checks

Compare assumptions and cost drivers.

Limited by assumed operation

Annual averages hide hourly demand, networks and storage requirements.

MACRO evaluates the whole system

Capacity and dispatch respond to demand and system constraints.

Source: sensitivity to slide 7 assumptions. Same efficiency and unit variable costs. MACRO TEA appendix.

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